

Four published results

MAR

NeurIPS 2024

A per-token diffusion head replaces the softmax. No vector quantisation anywhere.

we take: the adaLN flow head

FLOAT

ICCV 2025

Flow matching in a 512-d motion latent beats Hallo on one A100, 22 days, 11.3 h of data.

we take: predict motion, not pixels

Teller

CVPR 2025

A streaming autoregressive model over motion reaches Sync-C 7.70 against real video's 8.09.

we take: block-causal chunks

LivePortrait

2024 · MIT licence

A frozen warping renderer already gives 32.01 dB and CSIM 0.913 at 256 px.

we take: the pixels, for free

SANG-M — streaming, block-causal, audio-conditioned motion-latent flow matching

one token = one frame's 70-d LivePortrait motion · one chunk = 4 frames · the renderer never trains · 1× H100, ~80 M parameters

Resampling Forcing

Teacher-free exposure-bias fix: self-resampled history, LogitNormal(0,1), shift 0.6.

we take: the whole recipe

EMO · VASA-1

A future-audio window, and classifier-free guidance on the audio condition.

we take: 4-tick lookahead, CFG 2

LatentSync · StableSync

A SyncNet loss belongs in stage two, with the backbone frozen.

lesson: ours was being gamed

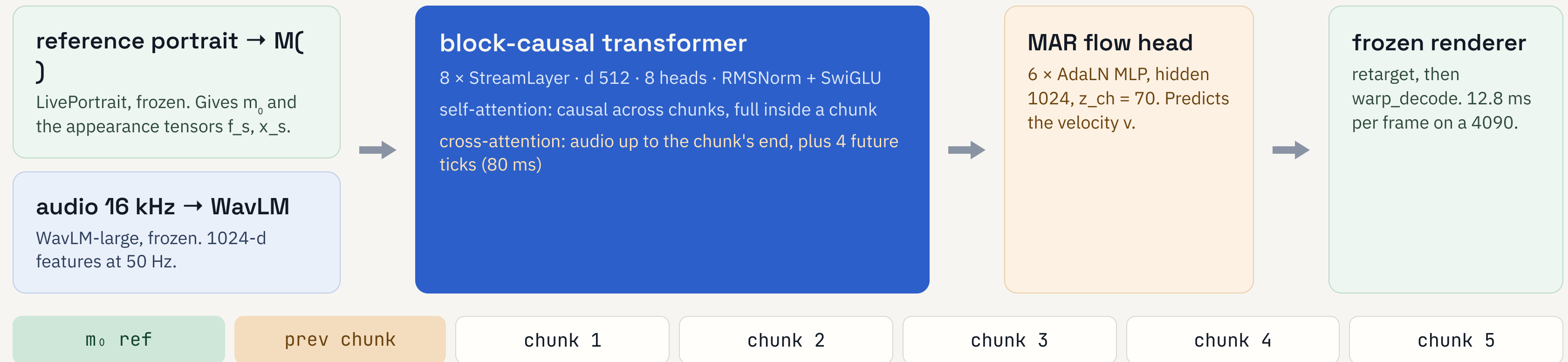
TalkVid

1,244 h, 7,729 speakers, and a 500-clip bench stratified by language, ethnicity, age.

we take: the fairness axis

VidTok-FSQ caps at **25.31 dB** @128 px over 300 clips; the Wan2.1 VAE reaches **31.80 dB** @256 px; and the appearance model still plateaued at **15.0 dB**, below the 16.25 dB freeze-context copy. Fixing the codec did not fix the task.

Audio in, 70 numbers per frame out, a frozen renderer for the rest



2 conditioning slices + 5 content chunks $\times$ 4 frames = 28 tokens. The appearance track needed 7,168 for a shorter window.

Training step

- 1 $\cdot$ Teacher forcing for the first 10% of steps, so the model becomes autoregressive at all.
- 2 $\cdot$ Then degrade the history: $t_s \sim \text{LogitNormal}(0,1)$ shifted by $s = 0.6$, one Euler step, no gradient. The context becomes the model's own imperfect output.
- 3 $\cdot$ Forward on that degraded context. Drop audio 0.1, reference 0.1, previous chunk 0.5.
- 4 $\cdot$ AdamW $3e-4 \rightarrow 3e-5$ cosine, wd 0.1, clip 1.0, bf16, batch 256, 200k steps $\approx$ 6–8 h on one H100.

The objective

$$L = L_{\text{flow}} + \lambda_{\text{vel}} \cdot L_{\text{vel}}$$

$$L_{\text{flow}} = E[\| v(z_t, t, h) - (\varepsilon - \Delta m) \|^2]$$

$z_t = (1-t) \cdot \Delta m + t \cdot \varepsilon$, $t \sim U(0,1)$, $h = \text{backbone}(\text{degraded history, audio})$

$$L_{\text{vel}} = \| (x[i+1] - x[i]) - (\Delta m[i+1] - \Delta m[i]) \|^2$$

$\hat{x} = z_t - t \cdot v$, the one-step clean estimate. $\lambda_{\text{vel}} = 1$, FLOAT's smoothness term.

Targets are always the clean Δm . Only the context is

HDTF	predicts	FID ↓	FVD ↓	Sync-C ↑	FPS ↑
EchoMimic	appearance	9.00	155.71	3.56	0.8
SoulX-FlashHead Pro	appearance	9.97	111.38	5.73	10.8
SoulX-FlashHead Lite	appearance	11.37	126.52	4.21	96
Ditto	motion	12.35	199.13	3.57	45.0
Sonic	appearance	13.53	113.31	5.17	offline
Hallo3	appearance	15.95	160.94	3.18	0.16
Ours · SANG-M	motion	10.17	115.38	4.41	25+ target

Protocol: SoulX-FlashHead Tab. 3 · HDTF, 75 clips · FPS on one H20, trained on 32× H20 over 782 h. Real video is Sync-C 8.09 on Teller's split.

The single-GPU frontier

FLOAT · 1× A100, 22 days, 11.3 h
FID 21.10 · FVD 162.05 · LSE-C 8.222
KDTalker · 43 M params, one RTX 4090
LSE-C 7.33 · FID 9.76 · 21.7 FPS

The protocol trap

SadTalker HDTF FID: **21.58** in SoulX's table, **71.95** in FLOAT's.
Hallo3 FID: **15.95** here, **871** in LeapTalk's.
Nothing crosses tables. Every claim needs one harness with the baselines re-run.

Sept 2026: convergence

Motar · 77 M, 15.4 FPS, ΔSync-C 0.08
AVTR-1 · 153 M, 2.81× real time, L40S
LeapTalk · 1 NFE, Sync-C 8.38, 200 FPS

Audio-driven one-shot talking heads: reported metrics

Paper	Main approach	Params	Metrics (dataset · values)
FLOAT	flow matching, motion latent	n/r	HDTF · LSE-C 8.22 · LSE-D 7.29 · FID 21.10 · CSIM 0.84
KDTalker	diffusion, LP keypoints	42.9 M + LP	HDTF · LSE-C 7.33 · LSE-D 7.55 · FID 9.76 · CSIM 0.95
Ditto	DiT on LP motion, real time	n/r + LP	HDTF† · Sync-C 3.57 · Sync-D 10.49 · FID 12.35
Teller	autoregressive motion tokens	n/r	HDTF · Sync-C 7.70 · Sync-D 7.54 · FID 21.35
Playmate	DiT in LP keypoint space	n/r + LP	own set · Sync-C 8.14 · Sync-D 7.06
Xemo-Talker	diffusion, 70-d + emotion	115.7 M + LP	MEAD · LSE-C 6.37 · LSE-D 8.21 · emo 85.3 %
Motar	AR + MLP diffusion head	77 M + 1.7 B	MEAD · FID 45.0 (real-motion oracle 44.2)
IMTalker	implicit-motion renderer	39 M + 124 M	HDTF recon · PSNR 28.46 · FID 7.43 (LP 9.05)
Hallo	pixel diffusion, SD-1.5	≈ 1.7 B est.	HDTF‡ · LSE-C 7.58 · LSE-D 7.79 · FID 25.36
EchoMimic	pixel diffusion, SD-1.5	≈ 1.7 B est.	HDTF‡ · LSE-C 6.24 · LSE-D 8.90 · FID 33.55
SadTalker	3DMM + warping renderer	n/r	HDTF‡ · LSE-C 7.31 · LSE-D 7.95 · FID 71.95
SoulX-FlashHead	Wan2.1 video DiT	1.3 B	HDTF · Sync-C 5.73 · Sync-D 8.77 · FID 9.97
SANG-M (ours)	flow-matching DiT, 42-d	53.4 M + LP	TalkVid · audio gain 1.50 · CSIM* 0.91 · HDTF pending

green: motion space · purple: pixel space · † SoulX Tab. 3 · ‡ FLOAT Tab. 1 · * renderer ceiling · real HDTF: LSE-C 8.24, LSE-D 6.93
Params = trainable generator · LP = frozen LivePortrait ≈ 130 M · n/r = not reported · est. = two SD-1.5 UNets (0.86 B each)

A prefix-slot mismatch between training and inference

1 slot = 1 video frame: its 42 motion numbers + its audio. The model reads 74 slots at once.

slots 0-9 slots 10-73

training: 10 context frames (or 10 blanks, half the time), then 64 to generate

context

64 frames to generate · loss here

old first window: no context slots, so frames 0-9 sat in slots 0-9

gen 0-9

gen 10-63

no slot

fix, --start null (default): 10 blanks, exactly as in training

blank ×10

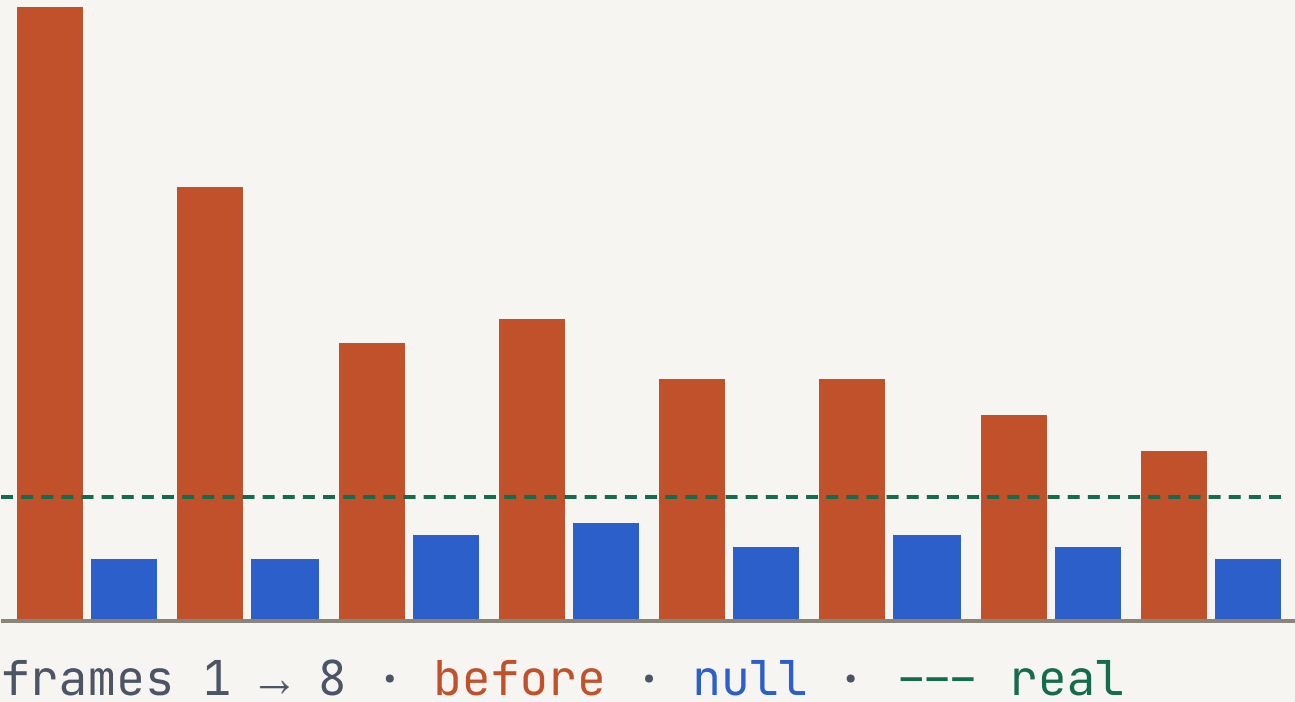
64 frames to generate

fix, --start source: the photo's pose ×10, with 0.4 s of silence

photo ×10

64 frames to generate

pixel acceleration, generated / real



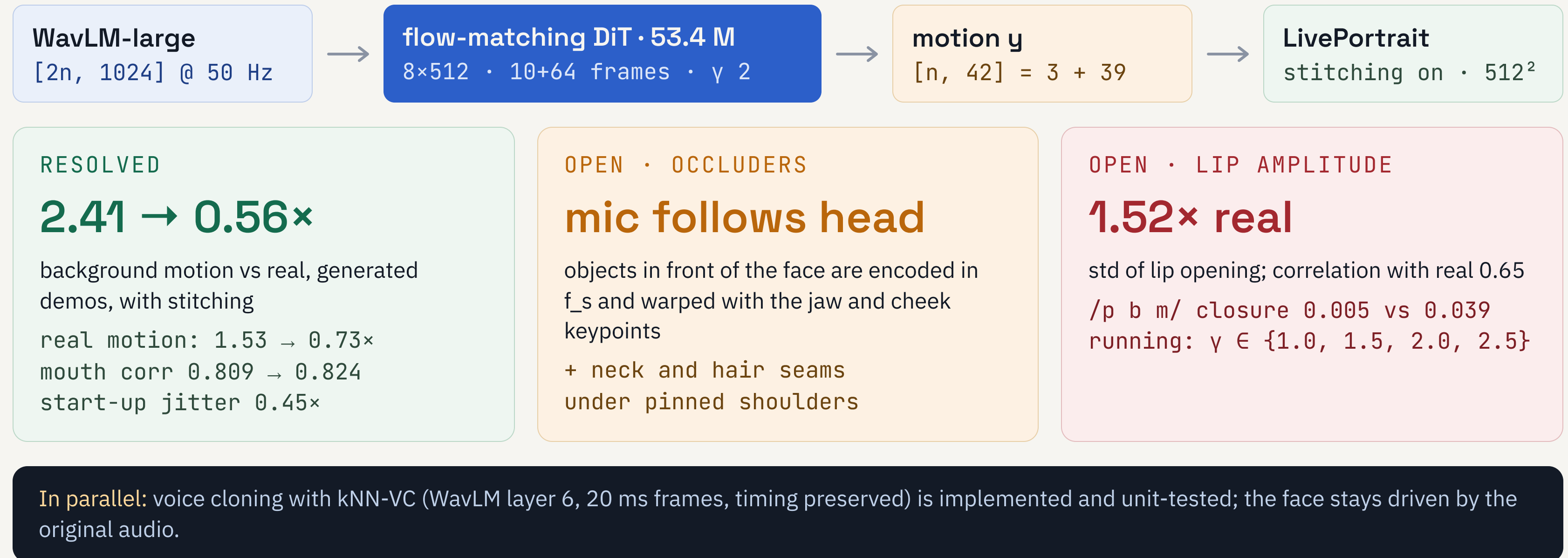
2.70 → 0.65

frames 0-10, ×real

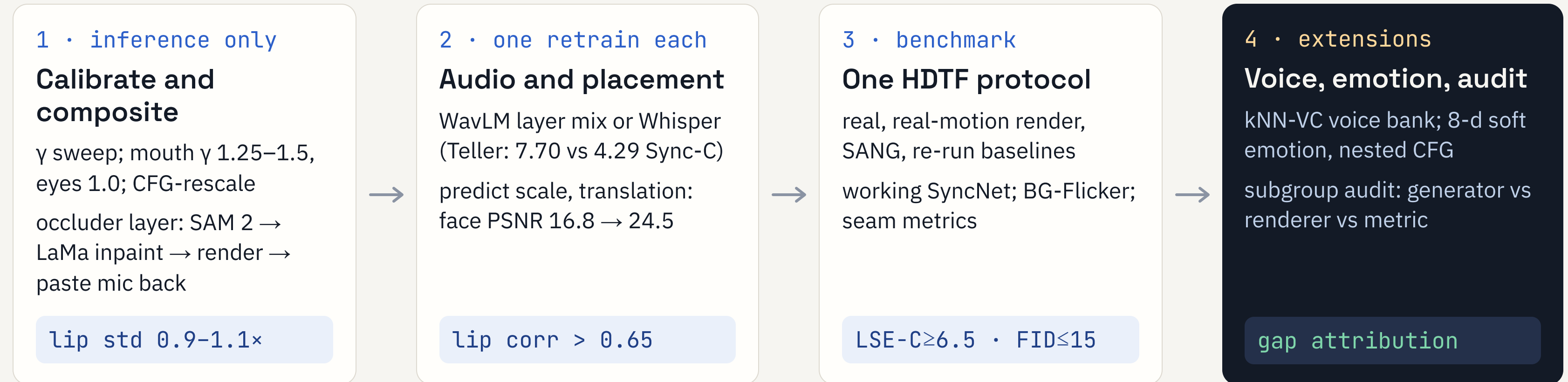
0

retraining steps

Stable background; occluders and lip amplitude remain



Four stages, each gated by a measurement



Research question. Where does motion-space generation fail across languages, ages and ethnicities, and is the loss in the generator, the renderer, or SyncNet itself?